

CAD Design Documentation Project Report

Mechanical Engineering Project

Abstract

This project focuses on the design and documentation of a mechanical component using Computer-Aided Design (CAD) software. The project includes 2D drawings, 3D modeling, material selection, manufacturing considerations, and engineering documentation.

Objectives

1. Create accurate 2D drawings.
2. Develop a 3D CAD model.
3. Generate engineering documentation.
4. Study materials and manufacturing processes.

Software Used

AutoCAD, SolidWorks, Fusion 360 (any CAD software).

Design Procedure

Select a component, create sketches, develop a 3D model, assemble components if required, generate engineering drawings, and prepare technical documentation.

Material Selection

Materials are selected based on strength, durability, cost, and manufacturing requirements.

Manufacturing Process

Typical processes include casting, machining, drilling, grinding, and assembly.

Applications

Mechanical design, manufacturing industries, product development, and engineering education.

Conclusion

The project demonstrates the use of CAD tools to improve design accuracy, visualization, and engineering communication while preparing professional engineering documentation.